بَيان Bayān — System Overview

Evidence-grounded Qur'an & Tafsir research assistant
Technical summary · September 2026

Bayān answers questions about the Qur'an in **Arabic or English** and shows the verses and tafsir it relied on. Its core rule: **no sentence reaches the reader unless it points to a real source that is displayed next to it.** Language models are used for understanding and wording; **code** decides what counts as evidence and renders every citation.

The model picks evidence. Code writes the citation.
Models return evidence IDs, never verse text or references — so a verse cannot be invented.

Every claim is checked.
A validator rejects unknown IDs and drops any sentence that adds ideas absent from its sources.

Honest when evidence is thin.
If nothing survives validation, the reader sees "insufficient evidence" and the sources — not a confident guess.

1 How a question becomes an answer

one turn, typically 20–45 s, every stage streamed live to the UI

Deterministic code · Cloud LLM (Groq) · Local LLM (on-device) · Database search

1	**Ask** Next.js UI	Arabic (RTL) or English question in a conversation thread. Progress arrives as a live **execution trace** over server-sent events.	Browser → API
2	**Scope check** precheck	Is this Qur'an / tafsir research? Out-of-scope requests (e.g. rulings, hadith) stop here with an explanation.	Rules
3	**Plan** understand the question	Turns the question into a **typed research plan**: intent, operations and search queries, returned as strict JSON and schema-validated.	gpt-oss-20b · Groq
4	**Retrieve** three search arms	**Lexical** — Arabic-normalised full-text + trigram. **Dense** — meaning search with bge-m3 embeddings (1024-d, HNSW). **Tafsir arm** — finds verses through their commentary.	PostgreSQL · pgvector bge-m3 · Cloudflare
5	**Fuse & hydrate** rank merge	**Reciprocal Rank Fusion** merges the arms by rank, not raw score. Top passages are loaded with their canonical Arabic text, translation and tafsir.	RRF
6	**Assemble evidence** + source rights	Each passage becomes an **evidence item** with an ID, a canonical pointer and a rights policy: may it be displayed, sent to a cloud model, or processed only locally?	Source policy
7	**Relevance** filter	A deterministic lexical + tafsir check drops passages that don't actually address the question.	Deterministic
8	**Synthesise** evidence-bound claims	The model writes short **claims, each citing evidence IDs**. If a rights-restricted tafsir is in the evidence (or Groq is rate-limited), this runs **on the local model** so the text never leaves the machine.	gpt-oss-20b · Groq or Gemma 4 · local
9	**Validate** GroundingValidator	Every cited ID must exist; Arabic/translation pairs are completed; an **entailment gate removes any claim** that introduces concepts absent from its evidence.	Deterministic gate
10	**Compose** reader-facing prose	Rewrites the **verified** claims into connected prose. It sees claims and reference labels only — not the source text — so it has nothing to paraphrase or invent.	gpt-oss-120b · Groq
11	**Render** deterministic citations	For each cited ID, code fetches the verse text, surah name and verse number from the database and draws the **citation chips and evidence cards**.	Code, not model

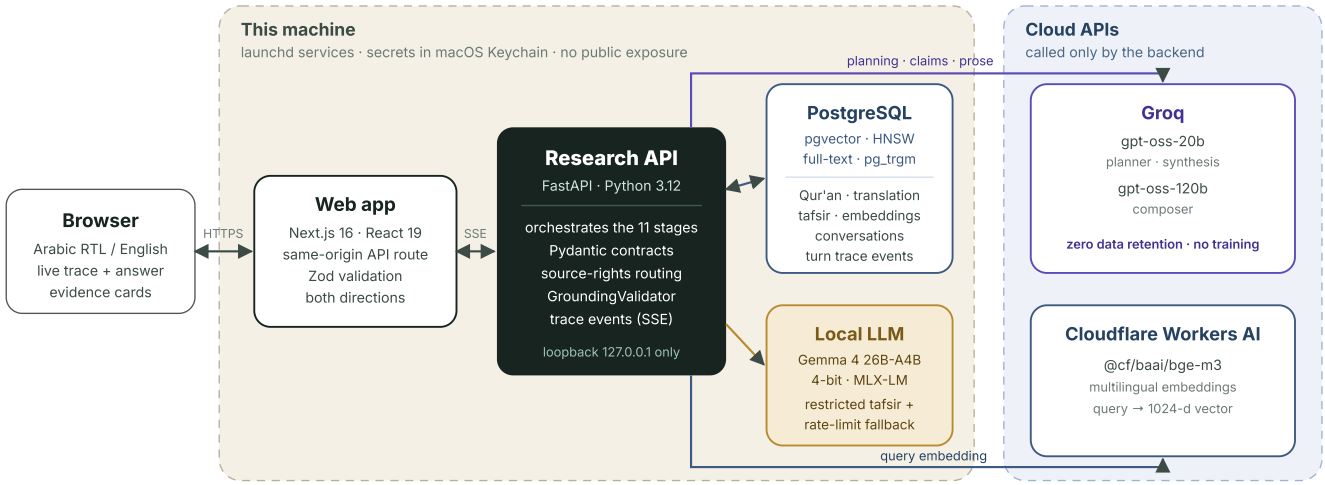
Honest fallback — if no claim survives step 9 (or the question has no support in the corpus), the turn ends as **"insufficient evidence"** and shows the retrieved sources, instead of producing a fluent but unsupported answer.

بَيان Architecture, models & tools

Where each part runs, and why

2 System architecture

local-first: the database, orchestrator and fallback model run on one machine



3 Models and why

Role	Model	Runs on	Why this choice
Planner	`gpt-oss-20b`	Groq	Fast, reliable strict-JSON plans
Synthesis	`gpt-oss-20b`	Groq	Evidence-bound claims; schema over style
Synthesis (local)	`Gemma 4 26B-A4B`	Local · MLX	Restricted tafsir stays on-device; Groq rate-limit fallback
Composer	`gpt-oss-120b`	Groq	The only reader-facing prose
Embeddings	`bge-m3`	Cloudflare	Arabic + English in one vector space
Relevance, validation	`no model`	Code	Auditable: same input, same decision

4 Sources

- **Qur'an (Arabic)** — Uthmani script, Tanzil Project
- **English translation** — Rowwad Translation Center (QuranEnc)
- **Tafsir** — al-Mukhtasar fi al-Tafsir (Tafsir Center for Quranic Studies), via the Quranic Universal Library
- Each source carries a **rights policy** that decides display, cloud use and local-only processing.
- English answers use Qur'an + translation only, until an approved English tafsir is available.

5 Trust by design

No invented citations
Models reference evidence IDs; verse text, surah names and numbers come from the database.

Claim-level validation
Unsupported sentences are removed individually; the rest of a good answer survives.

Honest fallback
"Insufficient evidence" plus sources, rather than a fluent guess.

Rights-aware routing
Source policy decides whether text may reach a cloud model or stays on-device.

Private providers
Groq route with zero data retention and no training; local model bound to loopback.

Observable
Every stage emits a trace event — streamed to the UI and stored per turn in Postgres.

6 Tooling & quality

FRONTEND
Next.js 16, React 19, Tailwind 4
Arabic-first RTL UI, Zod-validated contracts

BACKEND
FastAPI, Pydantic, httpx, psycopg
custom stage orchestrator, no agent framework

CONTRACTS
OpenAPI → TypeScript
type-check fails on any request/response drift

TESTING
376 backend tests (pytest)
+ Vitest unit and Playwright browser tests

Measured, not guessed. Retrieval and selection changes are evaluated on a set of hard, paraphrased questions before they are adopted — for example, a cross-encoder reranker was measured, made results worse, and was rejected.